Fully Dynamic Verifiable Secret Sharing and its Application to DPSS

Jiarui Li, Mengzhen Zou, Chen Qian and Guoyan Zhang

***Abstract*—Secret sharing serves as a critical primitive for distributed systems; however, traditional protocols lack the adaptability required for dynamic environments with unpredictable participant turnover. While Dynamic-Committee Proactive Secret Sharing (DPSS) attempts to address this, existing synchronous solutions suffer from a critical vulnerability: they necessitate a multi-round hand-off phase that effectively imposes a "quiescent window." This requirement forces committee membership to remain static during transitions, directly contradicting the premise of a fully dynamic adversary.**

In this work, we propose a Fully Dynamic-Committee model that eliminates the quiescent window by supporting committee reconfiguration in every round. We present two primary contributions. First, we introduce a suite of three perfectly secure Fully Dynamic Verifiable Secret Sharing (FDVSS) schemes that support per-round committee reconfiguration. Among them, our main construction, FDVSS-1, achieves near-optimal round complexity (four rounds) and polynomial communication ($O(n^4)$). Second, building on FDVSS, we construct a perfectly secure Fully Dynamic-Committee Proactive Secret Sharing (FDPSS) protocol. FDPSS achieves share redistribution in a single communication round, removing "quiescent window" during the hand-off phase. Our protocols require no trusted setup and provide optimal resilience ($t < n/3$) against a Byzantine adversary, offering a robust solution for fault-tolerant distributed systems in highly dynamic networks.

***Index Terms*—Verifiable secret sharing, dynamic committee, blockchain, distributed computing system**

I. Introduction

SECRET sharing [1] serves as the cornerstone of fault-tolerant distributed systems, ranging from secure multi-party computation (MPC) to threshold cryptography. While classical schemes assume a static set of participants, modern decentralized applications, such as permissionless blockchains and federated learning, operate in highly volatile environments characterized by constant node churn. In these settings, the static committee assumption collapses, rendering standard secret sharing insecure.

To address this, Proactive Secret Sharing (PSS) [2] and its extension, Dynamic-Committee PSS (DPSS) [3], [4], were developed to refresh shares across evolving committees. However, existing synchronous DPSS schemes suffer from significant limitations: they either incur prohibitive exponential communication costs [5], rely on strong trusted setups like KZG commitments [4], [6], or fail to achieve optimal Byzantine resilience [7], [8].

The "Quiescent Window" Vulnerability. Beyond efficiency, a more fundamental flaw exists in prior art. Existing schemes necessitate a multi-round hand-off phase, which inherently creates a "quiescent window", a period effectively requiring the committee to remain static. Crucially, this assumption imposes artificial stability constraints on the mobile adversary, thereby failing to capture the arbitrary nature of node churn in dynamic settings. This is not merely a theoretical concern but a practical reality in permissionless blockchains where node availability is unpredictable; participants cannot guarantee online presence for a predetermined interval or the entire duration of a hand-off.

Furthermore, prior work on PSS and DPSS has focused almost exclusively on the refresh phase, where the committee already holds shares.The initial distribution of the secret, typically realized via Verifiable Secret Sharing (VSS), mandates similar multi-round interaction and is equally susceptible to mobile adversaries, yet has been largely overlooked

Inspired by the recent YOSO (You Only Speak Once) [9] and Fluid MPC [10] paradigms, we advocate for a Fully Dynamic-Committee model. In this strict model, the participant set is permitted to change in every round, effectively eliminating the vulnerability of a multi-round static window. The central challenge we address is:

Is it possible to design perfectly secure vSS and DPSS schemes for fully dynamic committees that are efficient in both round complexity and communication?

A. Our Contributions

We provide an affirmative answer to the central research question by introducing two primary contributions: a suite of Fully Dynamic Verifiable Secret Sharing (FDVSS) schemes and a Fully Dynamic-Committee Proactive Secret Sharing (FDPSS) protocol. Both are designed for the synchronous model and achieve perfect security and optimal resilience against a mobile Byzantine adversary, where the adversary can corrupt at most t out of n participants, with $t < n/3$.

- **Fully Dynamic Verifiable Secret Sharing (FDVSS).** We introduce FDVSS, a verifiable secret sharing protocol that supports per-round committee reconfiguration. Our core contribution is a Shamir-based protocol (FDVSS-1).

Jiarui Li and Mengzhen Zou contributed equally to this work. (Corresponding author: Guoyan Zhang.)

Jiarui Li is with the School of Cyber Science and Technology, Shandong University, Qingdao, China (e-mail: 202421325@mail.sdu.edu.cn).

Mengzhen Zou is with the School of Cyber Science and Technology, Shandong University, Qingdao, China (e-mail: 202417038@mail.sdu.edu.cn).

Chen Qian is with the School of Cyber Science and Technology, Shandong University, Qingdao, China(e-mail: chen.qian@sdu.edu.cn).

Guoyan Zhang is with the School of Cryptologic Science and Engineering, Shandong University, Jinan, China(e-mail: guoyanzhang@sdu.edu.cn).

Crucially, it completes in four rounds, only one round above the known three-round lower bound for VSS, demonstrating near-optimal round complexity. It achieves a communication complexity of $O(n^4)$ under the cross-committee transfer assumption, and we further provide a variant (FDVSS-2) that achieves $O(n^5)$ without this assumption.

To build intuition, we begin by describing a simpler FDVSS protocol based on CNF-style secret sharing (FDVSS-0). Although it incurs exponential communication, its construction is more transparent and remains practically efficient for small committees, requiring only simple local computations.

- **Fully Dynamic-Committee Proactive Secret Sharing (FDPSS).** Building on FDVSS-1, we construct a perfectly secure FDPSS protocol with $O(n^5)$ communication complexity. This protocol directly addresses the critical limitations of prior art by supporting continuous reconfiguration during both the initial distribution and the redistribution phases. Crucially, it achieves share redistribution in a single communication round, thereby strictly eliminating the vulnerable "quiescent window" associated with static hand-off phases. Moreover, our construction achieves these properties without relying on any trusted setup.

B. Related Work

1) Synchronous VSS Schemes with Perfect Security: In the synchronous model, perfect security for VSS strictly mandates $n > 3t$ [11]. Ben-Or et al. (BGW) [12] first proposed a seven-round protocol satisfying this bound. Gennaro et al. (GIKR) [13] significantly optimized this result, presenting a four-round protocol with $O(n^2)$ communication and establishing the three-round lower bound. Subsequently, Katz and Koo [14] and Applebaum et al. [15] successfully achieved strict round optimality (three rounds) with $O(n^3)$ communication. Nevertheless, these foundational schemes are predicated on a static committee assumption, requiring participants to remain online and interactive throughout the multi-round execution. This limitation renders them inherently unsuitable for dynamic environments characterized by frequent node churn.

2) Synchronous DPSS Schemes: Proactive Secret Sharing [2] was originally introduced to defend against mobile adversaries by periodically refreshing shares within a static committee. Dynamic-Committee Proactive Secret Sharing extends this paradigm to facilitate committee reconfiguration. Wong et al. [5] proposed the first verifiable DPSS scheme; yet, it suffers from exponential communication complexity and relies on a "restart" mechanism upon fault detection, leading to unpredictable hand-off latency.

To address efficiency, Baron et al. [7] proposed a scheme achieving polynomial communication ($O(n^3)$) with statistical security. This efficiency, however, entails two significant tradeoffs. First, the protocol fails to achieve optimal resilience, tolerating only a $(1/2-\epsilon)n$ fraction of adversaries rather than the standard optimal threshold. Second, for the hand-off phase, despite guaranteeing a constant asymptotic round complexity ($O(1)$), the construction necessitates multiple invocations of complex sub-protocols (such as distributed matrix multiplication), resulting in a substantial number of concrete interaction rounds.

More recently, CHURP [4] leveraged dimension-switching for an optimistic 3-round hand-off. However, it relies on the KZG commitment scheme, which requires a trusted setup with a secret trapdoor. The exposure of this trapdoor would collapse the binding property of the commitment, thereby compromising security. Furthermore, in the presence of faults, CHURP reverts to a pessimistic path relying exclusively on on-chain communication. This introduces significant latency, as the dispute resolution process—involving on-chain share publication, verification, and accusation—requires multiple rounds of blockchain transactions.

A critical limitation shared by all these schemes is the necessity of a "quiescent window": they assume committee membership must remain static throughout the entire hand-off phase. This assumption creates a vulnerability in fully dynamic environments where nodes may churn at any round, which is a limitation our work specifically aims to eliminate. Table II provides a comparative summary of these representative DPSS schemes.

3) MPC and VSS with Dynamic Committees: Significant progress has been made in MPC for dynamic environments, spearheaded by the Fluid [10] and YOSO [9] paradigms. Subsequent research has extensively expanded these frameworks, addressing diverse challenges such as optimizing for linear communication [16], [17], removing trusted setup assumptions [18], [19], extending to asynchronous settings and relaxed fluidity models [20]–[22], or constructing robust random beacons [23], [24]. Despite these advancements, achieving Guaranteed Output Delivery (G.O.D.) remains expensive. Recent G.O.D. protocols [25], [26] typically incur high polynomial communication costs (e.g., $O(n^9)$ and $O(n^{18})$).

Dynamic VSS serves as the critical primitive for establishing G.O.D. in these frameworks, yet existing instantiations are inefficient. For instance, a statistically secure dynamic VSS was constructed within the YOSO framework [9]. However, this scheme serves primarily as a proof-of-concept; it necessitates a number of committees scaling polynomially with n and incurs prohibitive communication complexity. More relevant to our setting, David et al. [25] proposed a perfectly secure dynamic VSS. While achieving perfect security, their construction requires interaction across six distinct committees and incurs $O(n^6)$ communication complexity. Our DVSS protocol is specifically designed to optimize this foundation, significantly reducing both the committee overhead and communication costs. Table I provides a detailed comparison of representative synchronous VSS schemes under both static and dynamic membership models.

II. Preliminaries and Model

A. Fully Dynamic Model

We define the fully dynamic model, which operates in synchronous rounds. The model assumes secure point-to-point channels and a broadcast channel. The model is characterized by the following properties:

VSS scheme	Tolerance	Round/Committee	Security	Communication	Dynamic	Cross-committee Transfer Assumption
GIKR-VSS [13]	$n/3$	4	Perfect	$O(n^2)$	No	—
KKK-VSS [14]	$n/3$	3	Perfect	$O(n^3)$	No	—
YOSO-VSS [9]	$n/2$	$ploy(n)$	Statistical	$O(n^5)$	Yes	Yes
Layered-VSS [25]	$n/3$	6	Perfect	$O(n^6)$	Yes	No
Our FDVSS-0	$n/3$	4	Perfect	$exp(O(n))$	Yes	Yes
Our FDVSS-1	$n/3$	4	Perfect	$O(n^4)$	Yes	Yes
Our FDVSS-2	$n/3$	4	Perfect	$O(n^5)$	Yes	No

TABLE I: Comparison of Representative Synchronous VSS Schemes under Static (Perfect Security) and Dynamic Membership Models. For the "Cross-committee Transfer Assumption": '—' indicates a static setting (not applicable); 'Yes'/'No' indicates whether multi-committee message forwarding is assumed.

DPSS scheme	Tolerance	Security	Communication	Trusted Setup	Round of Hand-off	Quiescent Window
Wong et al. [5]	$n/2$	Computational	$exp(O(n))$	No	2 (Optimistic) Restart (Pessimistic)	Yes
Baron et al. [7]	$(1/2-\epsilon)n$	Statistical	$O(n^3)$	No	$O(1)$ (Worst-case)	Yes
CHURP [4]	$n/2$	Computational	$O(n^3)$	Yes	3 (Optimistic) Multi-round (Pessimistic)	Yes
Our FDPSS	$n/3$	Prefect	$O(n^5)$	No	1 (Worst-case)	No

TABLE II: Comparison of performance metrics of Synchronous DPSS Refresh. For the **Round of Hand-off**: 'Optimistic' and 'Pessimistic' refer to execution paths without and with detected faults, respectively; 'Worst-case' indicates the round complexity is invariant to adversarial behavior. Specifically, 'Restart' implies the protocol must abort and re-execute; '$O(1)$' denotes a constant number of rounds (specifically > 10 rounds, determined by the underlying secret sharing primitives); 'Multi-round' refers to a sequence of interactive recovery steps (specifically > 6 on-chain rounds, contingent on dispute resolution). **Quiescent Window** indicates whether the protocol necessitates a static committee assumption during the hand-off phase.

- **Dynamic Committees:** The protocol is executed by a sequence of committees. In any round i, the active committee is Com_i, composed of n participants. The committee composition is fully dynamic; Com_i and Com_j (for $j \neq i$) may be disjoint or overlapping. We define λ as the *forwarding horizon*. Participants in Com_i are aware of the identities of participants in future committees Com_{i+k} for $k \in \{1, \ldots, \lambda\}$ and can send messages to them via secure point-to-point channels or broadcast channel.
- **Adversary Model:** We assume a Byzantine adversary who can corrupt at most $t < n/3$ participants in any given committee (the standard threshold for perfect security). We consider a *non-retroactive adaptive adversary*. This implies that when the adversary corrupts a party, it learns that party's current internal state (e.g., secret shares, randomness) but does not gain access to any state from that party's participation in previous committees. This security guarantee requires participants to perform perfect state erasure immediately upon the conclusion of their committee tenure.

In this work, we instantiate this model with two configurations. We first consider the case of $\lambda = 3$ (our default setting), assuming parties can communicate with participants up to three rounds in the future. We also analyze the more restrictive case of $\lambda = 1$, which permits forwarding only to the immediate successor committee, thereby minimizing the requisite lookahead horizon.

B. Robust Secret Sharing

We begin by formally defining Robust Secret Sharing. We then review two foundational Robust Secret Sharing schemes.

Definition 1. *A (t, n) robust secret sharing scheme is a tuple of algorithms $\mathsf{SS} = (\mathsf{Share}, \mathsf{Reconst})$ that allows a dealer to distribute a secret m among n parties such that the following properties hold:*

- ***Privacy:*** *Any subset of at most t shares gives no information about the secret m.*
- ***Robustness:*** *The secret m is uniquely determined by all n shares, even if up to t of them are faulty.*

1) CNF secret sharing: A (t, n)-CNF robust secret sharing scheme is a tuple of algorithms $\mathsf{SS} = (\mathsf{Share}, \mathsf{Reconst})$, where the message space $\mathcal{M}$ is a group. Let M be an enumeration of all $\binom{n}{n-t}$ subsets of size $n - t$ such that

$$\mathsf{M} = |\mathcal{S}| = \binom{n}{n-t}$$

$$\mathcal{S} = \{S \subseteq [n] \mid |S| = n - t\} = (\mathcal{S}_1, \ldots, \mathcal{S}_{\mathsf{M}}).$$

The algorithms are defined as follows [27]:

- $\mathsf{Share}(\mathsf{m})$: Given a message $\mathsf{m} \in \mathcal{M}$:
  1. Samples $\{\mathsf{m}_j\}_{j \in \mathsf{M}} \in \mathcal{M}$ uniformly at random, such that $\mathsf{m} = \sum_{j \in [\mathsf{M}]} \mathsf{m}_j$.
  2. For each $i \in [n]$, sets the i-th share as $[\![\mathsf{m}]\!]_i = \{\mathsf{m}_j | j \in [\mathsf{M}], i \in \mathcal{S}_j\}$.
  3. Outputs the shares $([\![\mathsf{m}]\!]_1, \ldots, [\![\mathsf{m}]\!]_n)$.
- $\mathsf{Reconst}([\![\mathsf{m}]\!]_1, \ldots, [\![\mathsf{m}]\!]_n)$:

1. For each $i \in [n]$ and $j \in \mathsf{M}$ such that $i \in \mathcal{S}_j$, denote the additive share m_j from $[\![\mathsf{m}]\!]_i$ as m_j^i. For each $j \in \mathsf{M}$, choose the unique value $\hat{\mathsf{m}}_j$ based on a majority voting rule, such that $|\{i \in [n] : \mathsf{m}_j^i = \hat{\mathsf{m}}_j\}| \geq n - 2t$.
2. Outputs $\mathsf{m} \leftarrow \sum_{j \in \mathsf{M}} \hat{\mathsf{m}}_j$.

2) Shamir secret sharing: A (t, n) Shamir robust secret sharing scheme is a tuple of algorithms $\mathsf{SS} = (\mathsf{Share}, \mathsf{Reconst})$, where the message space $\mathcal{M}$ is a finite field and $n \geq 3t + 1$. The algorithms are defined as follows [28]:

- $\mathsf{Share}(\mathsf{m})$: Given a message $\mathsf{m} \in \mathcal{M}$:
  1. Samples a polynomial $\mathsf{F} \in \mathcal{M}[X]$ of degree t uniformly at random, such that $\mathsf{F}(0) = \mathsf{m}$.
  2. For each $i \in [n]$, sets the i-th share as $[\![\mathsf{m}]\!]_i = \mathsf{F}(i)$.
  3. Outputs the shares $([\![\mathsf{m}]\!]_1, \ldots, [\![\mathsf{m}]\!]_n)$.
- $\mathsf{Reconst}([\![\mathsf{m}]\!]_1, \ldots, [\![\mathsf{m}]\!]_n)$:
  1. Using the shares, runs the Berlekamp-Welch algorithm [29] to recover a polynomial F of degree t that agrees with at least $n - t \geq 2t + 1$ shares.
  2. If such a polynomial does not exist, outputs $\perp$. Otherwise, outputs $\mathsf{m} \leftarrow \mathsf{F}(0)$.

C. Verifiable Secret Sharing

We begin by formally defining Verifiable Secret Sharing. We then review two foundational VSS protocols that serve as the basis for our work. In Sec.III, we will build upon these protocols to construct our FDVSS schemes.

Definition 2. A (t, n) verifiable secret sharing scheme is a protocol between a dealer D and n parties $\mathsf{P}_1, \ldots, \mathsf{P}_n$. The protocol consists of two phases:

- ***Sharing phase.*** *The dealer D, holding a secret m, interacts with the n parties over several communication rounds. At the end of this phase, each honest party P_i holds a secret share and associated verification information.*
- ***Reconstruction phase.*** *Each party P_i provides its share. A public reconstruction function is applied to these shares to recover the secret.*

The scheme satisfies the following properties:

- ***Correctness.*** *If D is honest, the value reconstructed in the reconstruction phase is always equal to the original secret m.*
- ***Privacy.*** *If D is honest, the adversary's view during the sharing phase reveals no information about m.*
- ***Commitment.*** *Even if D is dishonest, a unique value m^* is determined at the end of the sharing phase, and m^* will be reconstructed in the reconstruction phase.*

1) CNF VSS: Under the condition of $n > 3t$, a VSS scheme based on CNF robust secret sharing is proposed in [13]. The scheme requires three rounds of communication. Note that the number of rounds in VSS refers to the communication rounds during the sharing phase, while reconstruction typically requires only one round.

- Sharing phase:
  1. Round 1(Distribute). The dealer samples additive shares $\mathsf{m}_1, \ldots, \mathsf{m}_\mathsf{M}$ such that $\sum_{i \in [\mathsf{M}]} \mathsf{m}_i = \mathsf{m}$, sends m_i to each party in $\mathcal{S}_i$. In $\mathcal{S}_i$, each party P_l sends a uniform random mask $r_{l,l'}$ to each other party $\mathsf{P}_{l'}$.
  2. Round 2(Broadcast and compare). For each pair $(\mathsf{P}_l, \mathsf{P}_{l'})$ in $\mathcal{S}_i$, the parties verify that the values m_i received from the dealer are equal. To do so, they broadcast an additively masked version of the secret $\mathsf{m}_i + r_{l,l'}$. If the two broadcasts differ, a complaint is marked against $\mathcal{S}_i$.
  3. Round 3(Correct). If a complaint is made against $\mathcal{S}_i$, this indicates that parties may be holding different values of m_i. The dealer needs to publicly broadcast m_i.
- Reconstruct phase: For each $\mathcal{S}_i$, if a complaint exists, parties in $\mathcal{S}_i$ broadcast m_i provided by the dealer in Round 3; otherwise, they broadcast the m_i they received in Round 1. Run the $\mathsf{Reconst}$ algorithm to recover m.

This protocol ensures privacy because the adversary, corrupting at most t parties, cannot learn all shares because there exists at least one all-honest set $\mathcal{S}_i$. Correctness and Commitment are guaranteed by the "compare-and-correct" mechanism: the broadcast in Round 2 ensures all honest parties in each set detect any inconsistency, and the dealer's correction in Round 3 forces commitment to a single value for that set.

2) Shamir VSS: An efficient four-round Shamir VSS protocol is also proposed in [13] with $n > 3t$.

- Sharing phase:
  1. Round 1(Distribute). The dealer samples a bivariate $\mathsf{F}(x, y)$ of degree (at most) t in each variable, such that $\mathsf{F}(0, 0) = \mathsf{m}$. The dealer sends $f_i(x) = \mathsf{F}(x, i)$ and $g_i(y) = \mathsf{F}(i, y)$ to P_i. Each party P_i sends a uniform random mask $r_{i,j}$ to each P_j.
  2. Round 2(Broadcast and compare). Each P_i broadcasts the masked value $f_i(j) + r_{ij}$ and $g_i(j) + r_{ji}$ for all j. Parties check the values for consistency, and a mismatch in the (i, j) cross-check flags the pair as conflicting.
  3. Round 3(Fault Detection). For each conflict pair (i, j), $\mathsf{P}_i, \mathsf{P}_j$ and the dealer broadcast their respective values: $f_i(j)$, $g_j(i)$, and $F(j, i)$. Any party whose broadcast values disagrees with the dealer's is marked *unhappy*. If more than t parties become unhappy, disqualify the dealer.
  4. Round 4(Correct). For each unhappy P_i, the dealer broadcasts the polynomial $f_i(x)$. Every happy party P_j broadcasts $g_j(i)$ to publicly verify $f_i(x)$ by checking if $g_j(i) = f_i(j)$. If fewer than $2t + 1$ happy parties confirm this equality, disqualify the dealer.
- Reconstruct phase: Happy parties broadcast $f_i(0)$ received in round 1; unhappy ones broadcast $f_i(0)$ received in round 4. Run the Berlekamp-Welch algorithm on $(f_1(0), \ldots, f_n(0))$ to recover m.

This protocol satisfies the VSS properties and a proof sketch for these properties is provided in [13].

III. Fully Dynamic Verifiable Secret Sharing

In this section, we present fully dynamic verifiable secret sharing schemes, with instantiations based on CNF secret

sharing and Shamir secret sharing. We begin with the CNF-based FDVSS (FDVSS-0), which follows the same high-level idea as the Shamir-based FDVSS (FDVSS-1) but has a simpler and more transparent structure. FDVSS-1 achieves better communication complexity at the cost of a more intricate design. Lastly, we propose two primitives that replace the cross-committee messaging components of FDVSS-1, resulting in an FDVSS-2 protocol with $\lambda = 1$.

A. FDVSS-0

Our FDVSS-0 protocol builds upon the static CNF VSS scheme from section II-C1. However, the static protocol cannot be directly applied to the fully dynamic setting, as it relies on multi-round interactive complaints, but in the dynamic model, an adversary can corrupt different sets of parties in successive rounds. This allows them to accumulate more information than a static adversary, a vulnerability we term cross-committee corruption.

s Specifically, consider a native transplant for the CNF VSS protocol. An honest party $\mathsf{P}_l^0 \in \mathsf{Com}_0$ must first send the mask $r_{l,l'}$ to both its delegate $\mathsf{P}_l^1 \in \mathsf{Com}_1$ and $\mathsf{P}_{l'}^1 \in \mathsf{Com}_1$. However, we cannot guarantee that P_l^1 is also honest, and the discontinuity in honest participation compromises the share's privacy. Similarly, in the static setting, the same party P_l uses the same local share in both Round 2 and the reconstruction phase. In the fully dynamic setting, an adversary leveraging cross-committee corruption could accumulate enough shares to reconstruct the secret, even if the set of dishonest parties within any single committee remains below the threshold. Moreover, the dealer would require a second round of messages to resolve share discrepancies, which violates the fully dynamic model.

To solve the cross-committee corruption problem, our key technique is dimension lifting. The dealer upgrades the one-dimensional share array $\{\mathsf{m}_i\}_{i\in[\mathsf{M}]}$ to a two-dimensional matrix $\{\mathsf{m}_{i,j}\}_{i,j\in[\mathsf{M}]}$, letting the sharing phase and the reconstruction phase operate along distinct indices to prevent cross-committee corruption. As shown in fig. 1, each set in the shareholder committee of the sharing phase holds shares along a distinct row, while each set in the shareholder committee of the reconstruction phase holds shares along a distinct column. Consequently, there exist fully honest sets $\mathcal{S}_i$ and $\mathcal{S}_j$ in the two committees, whose intersection share $\mathsf{m}_{i,j}$ remains unknown to the adversary, ensuring that m is still hidden.

Moreover, during interactive verification of shares among shareholders, the standard masking-based equality test in round 2 is not applicable in the fully dynamic setting. We delegate the test to another committee. To compare two values m_1 and m_2 held by two parties in Com_1, they share the values with an auxiliary committee, Com_2. Parties in Com_2 then compute and broadcast the share-wise difference, enabling a public reconstruction of $\mathsf{m}_1 - \mathsf{m}_2$ to verify if it is equal to zero.

Finally, to replace the dealer's correction round, the dealer pre-shares all individual shares with a dedicated auxiliary committee. If a discrepancy is detected, the parties in this committee collectively broadcast their shares of the disputed value. This public reconstruction simulates the original dealer's correction broadcast without requiring its continued availability.

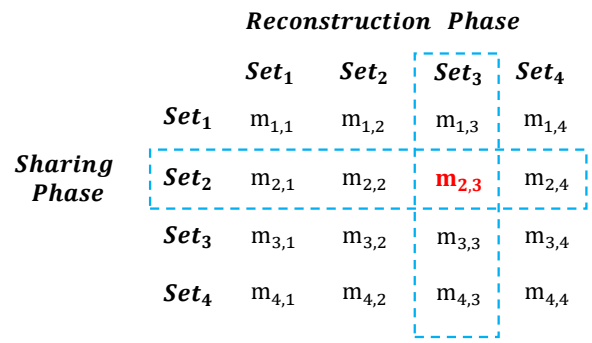

Fig. 1: An example of dimension lifting: The Sharing Phase operates on rows (e.g., all parties in $\mathcal{S}_2$ are honest) and the Reconstruction Phase operates on columns (e.g., all parties in $\mathcal{S}_3$ are honest). An all-honest row and column ensure their intersection share (e.g., $\mathsf{m}_{2,3}$) remains hidden from the adversary.

The details of the protocol are illustrated in fig. 2. We provide the security analysis and complexity analysis below.

1) Security Analysis:

Theorem 1. *The FDVSS-0 protocol constitutes a perfectly secure verifiable secret sharing scheme, satisfying the properties of correctness, privacy, and commitment against a Byzantine adversary corrupting up to $t < n/3$ parties in each committee.*

Proof. Correctness: Since the dealer D is honest, for each $i, j \in [\mathsf{M}]$, it sends $\mathsf{m}_{i,j}$ to $\mathcal{S}_i$ in Com_1 and shares $\mathsf{m}_{i,j}$ with Com_3. In Com_1, all parties in $\mathcal{S}_i$ hold a correct and consistent $\mathsf{m}_{i,j}$. If they execute the protocol correctly, for each pair ($\mathsf{m}_{i,j}^l$, $\mathsf{m}_{i,j}^{l'}$), the value reconstructed using the difference of shares broadcast by Com_2 must be 0, which is guaranteed by the robustness of CNF secret sharing. If a dishonest party P_l^1 in $\mathcal{S}_i$ shares a wrong $\mathsf{m}_{i,j}$, the difference between it and an honest party $\mathsf{P}_{l'}^1$ will not be 0, making $\mathsf{m}_{i,j}$ disputed and triggering Com_3 to broadcast the shares of $\mathsf{m}_{i,j}$ from D. And the value $\mathsf{m}_{i,j}^{\mathsf{D}}$ reconstructed from these shares is also guaranteed to be correct. For the shareholder committee Com_4, since each undisputed $\mathsf{m}_{i,j}$ is held by at least $t+1$ honest parties in Com_1 and correctly sent to $\mathcal{S}_j$ of Com_4, all parties in $\mathcal{S}_j$ of Com_4 will obtain the correct $\mathsf{m}_{i,j}$ through a majority vote. For each disputed $\mathsf{m}_{i,j}$, the reconstructed $\mathsf{m}_{i,j}^{\mathsf{D}}$ is used instead. At this point, each $\mathcal{S}_j$ in Com_4 obtains the correct $\mathsf{m}_{1,j}, \ldots, \mathsf{m}_{\mathsf{M},j}$ to compute m_j. In the reconstruction phase, since there are at least $t+1$ honest parties in each set, $\mathsf{m} = \sum_{j\in[\mathsf{M}]} \mathsf{m}_j$ can certainly be reconstructed.

Privacy: When D is honest, since the adversary can corrupt at most t parties in each committee, there must be at least one all-honest set in Com_1, denoted by $\mathcal{S}_i$ and at least one all-honest set in Com_4, denoted by $\mathcal{S}_j$. $\mathcal{S}_i$ in Com_1 hold $\{\mathsf{m}_{i,j}\}_{j\in[\mathsf{M}]}$, while $\mathcal{S}_j$ in Com_4 hold $\{\mathsf{m}_{i,j}\}_{i\in[\mathsf{M}]}$. Consequently, $\mathsf{m}_{i,j}$ remains hidden from the adversary. Furthermore, since D and all parties of $\mathcal{S}_i$ in Com_1 independently and randomly share $\mathsf{m}_{i,j}$, the adversary corrupting t parties in Com_2 and Com_3 is unable to reconstruct $\mathsf{m}_{i,j}$ even if it obtains

Protocol $\Pi_{\mathsf{FDVSS-0}}$

Dealer $\mathsf{D} \in \mathsf{Com}_0$**:**

1) Sample random values $\{\mathsf{m}_{i,j}\}_{i,j\in[\mathsf{M}]}$ such that $\mathsf{m} = \sum_{i,j\in[\mathsf{M}]} \mathsf{m}_{i,j}$.
2) For each $i \in [\mathsf{M}]$, send $\{\mathsf{m}_{i,j}\}_{j\in[\mathsf{M}]}$ to all parties P_l^1 where $l \in \mathcal{S}_i$.
3) For each $i, j \in [\mathsf{M}]$, sample $(\llbracket \mathsf{m}_{i,j} \rrbracket_1, \llbracket \mathsf{m}_{i,j} \rrbracket_2, \ldots, \llbracket \mathsf{m}_{i,j} \rrbracket_n) \leftarrow \mathsf{CNF.Share}(\mathsf{m}_{i,j})$ and send $\llbracket \mathsf{m}_{i,j} \rrbracket_k$ to P_k^3 for each $k \in [n]$.

Com 1: For each $i \in [\mathsf{M}]$ and $l \in \mathcal{S}_i$, P_l^1 proceeds as follows:

1) For each $j \in [\mathsf{M}]$, sample $(\llbracket \mathsf{m}_{i,j} \rrbracket_1, \llbracket \mathsf{m}_{i,j} \rrbracket_2, \ldots, \llbracket \mathsf{m}_{i,j} \rrbracket_n) \leftarrow \mathsf{CNF.Share}(\mathsf{m}_{i,j})$ and send $\llbracket \mathsf{m}_{i,j} \rrbracket_u$ to P_u^2 for each $u \in [n]$.
2) For each $j \in [\mathsf{M}]$, send $\mathsf{m}_{i,j}$ to each party P_l^4 where $l \in \mathcal{S}_j$.

Com 2: For each $u \in [n]$, P_u^2 proceeds as follows:

1) For each $i, j \in [\mathsf{M}]$ and each pair $l, l' \in \mathcal{S}_i, l \neq l'$, compute $\llbracket \mathsf{m}_{i,j} \rrbracket_u^l - \llbracket \mathsf{m}_{i,j} \rrbracket_u^{l'} = \{\mathsf{m}_{i,j,v}^l - \mathsf{m}_{i,j,v}^{l'} | s \in [\mathsf{M}], u \in \mathcal{S}_v\}$, broadcast $\llbracket \mathsf{m}_{i,j} \rrbracket_u^l - \llbracket \mathsf{m}_{i,j} \rrbracket_u^{l'}$. ($\llbracket \mathsf{m}_{i,j} \rrbracket_u^l$ means $\llbracket \mathsf{m}_{i,j} \rrbracket_u$ from P_l^1.)

Public Computation (Com 3): For each $i, j \in [\mathsf{M}]$ and each pair $l, l' \in \mathcal{S}_i, l \neq l'$, reconstruct $\mathsf{m}_{i,j}^l - \mathsf{m}_{i,j}^{l'} \leftarrow \mathsf{CNF.Reconst}(\{\llbracket \mathsf{m}_{i,j} \rrbracket_u^l - \llbracket \mathsf{m}_{i,j} \rrbracket_u^{l'}\}_{u\in[n]})$. If there exists any pair $l, l' \in \mathcal{S}_i, l \neq l'$ such that $\mathsf{m}_{i,j}^l - \mathsf{m}_{i,j}^{l'} \neq 0$, mark $\mathsf{m}_{i,j}$ as disputed.

Com 3: For each $k \in [n]$, P_k^3 proceeds as follows:

1) For each $i, j \in [\mathsf{M}]$, if $\mathsf{m}_{i,j}$ is disputed, broadcast $\llbracket \mathsf{m}_{i,j} \rrbracket_k^{\mathsf{D}}$ from D.

Public computation(Com 4): Reconstruct each $\mathsf{m}_{i,j} \leftarrow \mathsf{CNF.Reconst}(\{\llbracket \mathsf{m}_{i,j} \rrbracket_k\}_{k\in[n]})$. If the reconstruction fails, disqualify the dealer.

Com 4: For each $j \in [\mathsf{M}]$ and $l \in \mathcal{S}_j$, P_l^4 proceeds as follows:

1) For each $i \in [\mathsf{M}]$, set $\mathsf{m}_{i,j} = \mathsf{m}_{i,j}^{\mathsf{D}}$ if $\mathsf{m}_{i,j}$ has been disputed, otherwise set $\mathsf{m}_{i,j} = \mathsf{Maj}_{l'\in\mathcal{S}_i}(\mathsf{m}_{i,j}^{l'})$ from $\mathsf{P}_{l'}^1$ in $\mathcal{S}_i$ of Com_1.
2) The share of $\mathcal{S}_j$ is $\mathsf{m}_j = \sum_{i\in[\mathsf{M}]} \mathsf{m}_{i,j}$. If dealer is disqualified, set m_j to the default.

Fig. 2: FDVSS-0 Protocol

t shares of $\mathsf{m}_{i,j}$ from each of them. Additionally, the adversary cannot make $\mathsf{m}_{i,j}$ public through malicious behavior, as only the parties in Set_i of Com_1 can influence the consistency of $\mathsf{m}_{i,j}$. Therefore, nothing about m is revealed to the adversary.

Commitment: For each undisputed $\mathsf{m}_{i,j}$, the honest parties in $\mathcal{S}_i$ of Com_1 send the same $\mathsf{m}_{i,j}$ to $\mathcal{S}_j$ in Com_4. Regardless of what the dishonest parties in $\mathcal{S}_i$ send, all parties in $\mathcal{S}_j$ of Com_4 will obtain the same $\mathsf{m}_{i,j}$ through the majority vote. For each disputed $\mathsf{m}_{i,j}$, all parties in $\mathcal{S}_j$ of Com_4 set $\mathsf{m}_{i,j}$ to $\mathsf{m}_{i,j}^{\mathsf{D}}$, which is the value reconstructed using the shares broadcast by Com_3. At this point, for every $\mathcal{S}_j$ of Com_4, the honest parties in it hold a consistent share $\mathsf{m}_j^* = \mathsf{m}_{1,j} + \cdots + \mathsf{m}_{\mathsf{M},j}$. Therefore, if the dealer D is not disqualified, a value $\mathsf{m}^* = \mathsf{m}_1^* + \cdots + \mathsf{m}_{\mathsf{M}}^*$ can certainly be reconstructed whether D is honest or not. □

2) Complexity Analysis: The dominant communication overhead stems from the broadcast of Com_2. For each $\mathsf{m}_{i,j}$, all parties holding $\mathsf{m}_{i,j}$ in Com_1 split it into $\binom{n}{t}$ shares and it requires pairwise consistency checks between all parties holding $\mathsf{m}_{i,j}$ in Com_1, involving $O(n^2)$ pairs. Therefore, it leads to a total communication complexity of $O(n^2 \cdot \binom{n}{t}^3)$.

B. FDVSS-1

Our FDVSS-1 protocol is based on the static Shamir VSS scheme from section II-C2. When porting it to the fully dynamic setting, analogous issues to the CNF-based protocol arise: (i).The same pairwise masks are held by different parties in Round 1 and Round 2; (ii).The same shares are held by different parties in Round 2 and the reconstruction phase.

To address these issues, we employ dimension lifting again, elevating the standard bivariate polynomial $\mathsf{F}(X, Y)$ to a 4-dimensional form $\mathsf{F}(X, Y, Z, W)$. The additional dimensions serve distinct purposes: the Z dimension is used to create verifiable "slices" for the delegated comparison phase replacing the masks used in the static model, while the W dimension is reserved for the reconstruction phase. This separation ensures that the shares for verification in the Z dimension are independent of the final shares for reconstruction in the W dimension, solving the cross-committee corruption problem.

Although the mask-then-broadcast technique is infeasible in the dynamic setting, our protocol preserves the round efficiency of the underlying static VSS. Share exchange ($\mathsf{Com}_0 \rightarrow \mathsf{Com}_1$) and equality test ($\mathsf{Com}_1 \rightarrow \mathsf{Com}_2$) still require two rounds. However, by making the inconsistency detection public and part of the Com_2 comparison itself, we eliminate the separate "Fault Detection" round (Round 3) of the static protocol. Consequently, the overall round complexity remains four, matching the protocol in static setting.

The specific steps of the protocol are shown in fig. 3, and the diagram for message passing among the committees is shown

in fig. 4. We briefly describe the main process:

- Com_0: The Dealer selects a random degree-t, 4-dimensional polynomial $\mathsf{F}(X, Y, Z, W)$ such that $\mathsf{F}(0, 0, 0, 0) = \mathsf{m}$. It sends the 3-dimensional polynomials $\mathsf{F}(X, i, Z, W)$ and $\mathsf{F}(i, Y, Z, W)$ to each P_i^1 in Com_1. Simultaneously, it shares the "correction set" $\{\mathsf{F}(X, i, 0, W)\}_{i\in[n]}$ with Com_3 to enable delegated correction.
- Com_1: Upon receiving its data, each P_i^1 immediately sends relevant information to the future committees as follows:
  - For comparison (to Com_2), it sends the 2-dimensional slices $\mathsf{F}(X, i, j, W)$ and $\mathsf{F}(i, Y, j, W)$ (i.e., its 3-dimensional polynomials evaluated at $Z = j$) to the corresponding P_j^2.
  - For verification (to Com_3), it shares $\{\mathsf{F}(i', i, 0, W)\}_{i'\in[n]}$ with Com_3.
  - For reconstruction (to Com_4), it sends the 2-dimensional slice $\mathsf{F}(X, i, Z, k)$ to each P_k^4.
- Com_2: Each P_j^2 receives its slices $\{\mathsf{F}(X, i, j, W)\}_{i\in[n]}$ and $\{\mathsf{F}(i, Y, j, W)\}_{i\in[n]}$. It uses these to check all pairwise consistencies (e.g., verifying $\mathsf{F}(i', i, j, W) = \mathsf{F}(i, i', j, W)$) and broadcasts the resulting consistency graph G_j.
- Com_3: By aggregating the graphs $\{G_j\}_{j\in[n]}$, a global set of mutually consistent parties in Com_1 is publicly identified, as detailed in the Public Computation phase of Com_3 in fig. 3. For any party P_i^1 outside this set (i.e., inconsistent), Com_3 performs the delegated correction by broadcasting the shares of the polynomial $\mathsf{F}(X, i, 0, W)$ from dealer and the corresponding verification shares $\{\mathsf{F}(i', i, 0, W)\}_{i'\in[n]}$ from Com_1 for public verification.
- Com_4: Finally, each P_k^4 obtains a set of consistent values $\mathsf{F}(X, 1, 0, k), \ldots, \mathsf{F}(X, n, 0, k)$, and uses $\mathsf{F}(0, 1, 0, k), \ldots, \mathsf{F}(0, n, 0, k)$ to interpolate and compute $\mathsf{F}(0, 0, 0, k)$ as its share of m.

1) Security Analysis:

Theorem 2. *The FDVSS-1 protocol is a perfectly secure verifiable secret sharing scheme in the fully dynamic model. It satisfies the properties of correctness, commitment, and privacy against a Byzantine adversary that corrupts up to $t < n/3$ parties in each committee.*

Proof Sketch. For simplicity, we provide a sketch of the security proof here. A detailed proof is deferred to the Appendix.

Correctness. Define H_1 as the set of all the honest parties in Com_1, and H_2 as the set of all the honest parties in Com_2. If the dealer is honest, all parties in H_1 will definitely be happy because they use the correct shares sent by the dealer for comparison, and their pairwise consistency checks will pass. Therefore, H_1 must be present in the cliques formed by the consistency graphs of H_2, and they will be added to the set I. If a dishonest party P_i^1 in Com_1 uses the wrong shares to generate the slices for comparison, they will definitely be marked as unhappy because the cliques in H_2 include all honest parties in Com_1. For P_i^1 to be added to the clique, its pairwise consistency check with H_1 must pass, but this condition only holds when P_i^1 uses the shares from dealer for comparison. Moreover, due to the robustness of Shamir secret sharing, if dishonest P_i^1 uses the correct polynomial for comparison but sends the wrong $\mathsf{F}(X, i, 0, k)$ to P_k^4, it does not affect the correct reconstruction of $\mathsf{F}(0, 0, 0, k)$ because the slices received by P_k^4 from H_1 are all correct. Furthermore, even if a dishonest party in Com_2 broadcasts an wrong consistency graph, it does not affect the determination of I. Similarly, even if the dishonest parties in Com_3 broadcast wrong shares, it does not affect the reconstruction result. In the same way, each honest party P_k^4 in Com_4 will broadcast the correct share $\mathsf{F}(0, 0, 0, k)$ during the reconstruction phase, and even if other dishonest parties broadcast wrong shares, m will still be reconstructed.

Privacy. The secrecy of m is protected by the properties of the quadrivariate polynomial of degree t. The adversary controls at most t parties in any committee, thus obtaining at most t shares along any dimension. Since the polynomial has degree t in each variable, possession of t points is insufficient to reconstruct the polynomial or distinguish the value of the constant term (the secret m). The shares received by dishonest parties in Com_3 act as masking factors and provide no additional information about the secret due to the independence of the polynomial coefficients.

Commitment. If the dealer is not disqualified, this means that there are at least $2t + 1$ happy parties in Com_1, and define H_1' as the set of all the honest parties among them, where $|H_1'| \geq t + 1$. H_1' exists in the cliques formed by at least $2t + 1$ parties in Com_2, and define H_2' as the set of all the honest parties among them, where $|H_2'| \geq t + 1$. For each party $\mathsf{P}_j^2 \in H_2'$, the consistency check of all parties in H_1' at $Z = j$ passes, resulting the shares of H_1' come from a quadrivariate degree-t polynomial $G(X, Y, Z, W)$ according to the uniqueness of multivariate polynomial interpolation. For each unhappy party P_i^1, the polynomial $\mathsf{F}^D(x, i, 0, w)$ broadcast by the dealer satisfies $\mathsf{F}^D(i', i, 0, w) = \mathsf{F}^{i'}(i', i, 0, w)$ for at least $2t + 1$ happy parties (i.e., at least $t + 1$ parties P_i^1 in H_1'). Since $F^{i'}(i', i, 0, w) = G(i', i, 0, w)$ and $\mathsf{F}^D(x, i, 0, w)$ and $G(X, i, 0, w)$ are both degree-t in X, it follows that $\mathsf{F}^D(x, i, 0, w) = G(X, i, 0, w)$. Regardless of whether the values that dishonest but happy parties in Com_1 send to P_k^4 is correct or not, P_k^4 will definitely be able to reconstruct $G(0, 0, 0, k)$ using polynomials from other parties because $|H_1' \cup unhappy| \geq 2t + 1$. Therefore, in the reconstruction phase, a value s will definitely be reconstructed, where $s = G(0, 0, 0, 0)$.

□

2) Complexity Analysis: The Dealer constructs a 4-dimensional polynomial for m and distributes the corresponding shares to Com_1, which incurs an $O(n^4)$ communication overhead. The subsequent interactions between committees generate at most an additional $O(n^4)$ in communication costs. This leads to a total communication complexity of $O(n^4)$.

C. FDVSS-2

To achieve a maximum cross-committee forwarding distance of $\lambda = 1$, we derive FDVSS-2 from FDVSS-1 by

Protocol $\Pi_{\mathsf{FDVSS-1}}$

Dealer $\mathsf{D} \in \mathsf{Com}_0$**:**

1) Sample a random quadrivariate polynomial $\mathsf{F}(X, Y, Z, W)$ of degree t in each variable, s.t. $\mathsf{F}(0, 0, 0, 0) = \mathsf{m}$.
2) For each $i \in [n]$,
   - send $\mathsf{F}(i, Y, Z, W)$ and $\mathsf{F}(X, i, Z, W)$ to P_i^1.
   - sample $([\![\mathsf{F}(X, i, 0, W)]\!]_1, \ldots, [\![\mathsf{F}(X, i, 0, W)]\!]_n) \leftarrow \mathsf{Shamir.Share}(\mathsf{F}(X, i, 0, W))$, and send $[\![\mathsf{F}(X, i, 0, W)]\!]_u$ to P_u^3 for each $u \in [n]$.

Com 1: For each $i \in [n]$, P_i^1 proceeds as follows:

1) For each $j \in [n]$, send $\mathsf{F}(X, i, j, W)$ and $\mathsf{F}(i, Y, j, W)$ to P_j^2 in Com_2.
2) For each $i' \in [n]$, sample $([\![\mathsf{F}(i, i', 0, W)]\!]_1, \ldots, [\![\mathsf{F}(i, i', 0, W)]\!]_n) \leftarrow \mathsf{Shamir.Share}(\mathsf{F}(i, i', 0, W))$, send $[\![\mathsf{F}(i, i', 0, W)]\!]_u$ to P_u^3 for each $u \in [n]$.
3) For each $k \in [n]$, send $\mathsf{F}(X, i, 0, k)$ to P_k^4 in Com_4.

Com 2: For each $j \in [n]$, P_j^2 proceeds as follows:

1) For $i \in [n]$, it receives $\mathsf{F}^i(X, i, j, W)$ and $\mathsf{F}^i(i, Y, j, W)$ from P_i^1. For each pair of $i, i' \in [n]$, check $\mathsf{F}^i(i', i, j, W) = \mathsf{F}^{i'}(i', i, j, W)$ and $\mathsf{F}^i(i, i', j, W) = \mathsf{F}^{i'}(i, i', j, W)$. If both are met, P_i^1 and $\mathsf{P}_{i'}^1$ are considered consistent.
2) Define a consistency graph G_j on n nodes. If P_i^1 and $\mathsf{P}_{i'}^1$ are consistent, add an edge (i, i') to the graph.
3) Broadcast G_j.

Public computation(Com 3):

1) For each consistency graph G_j, identify the subgraph induced by vertices with a degree of at least $2t$. Check if this subgraph forms a clique of size at least $2t + 1$. If so, denote this clique as C_j. If fewer than $2t + 1$ such cliques are identified, disqualify the dealer.
2) Define a set of indices I, for each $i \in I$, i exists in at least $2t + 1$ C_j. If $|I| < 2t + 1$, disqualify the dealer.
3) Define a party set $\mathsf{Unhappy}$. For each $i \in [n]$, if $i \notin I$, mark $\mathsf{P}_i^1 \in \mathsf{Unhappy}$.

Com 3: For each $\mathsf{P}_i^1 \in \mathsf{Unhappy}$ and each $\mathsf{P}_{i'}^1 \notin \mathsf{Unhappy}$, each $\mathsf{P}_u^3 \in \mathsf{Com}_3$ proceeds as follows:

1) Broadcast $[\![\mathsf{F}^{\mathsf{D}}(X, i, 0, W)]\!]_u$ from D.
2) Broadcast $[\![\mathsf{F}^{i'}(i', i, 0, W)]\!]_u$ from $\mathsf{P}_{i'}^1$.
3) Broadcast the $\mathsf{Unhappy}$ set.

Public computation(Com 4): For each $\mathsf{P}_i^1 \in \mathsf{Unhappy}$ and each $\mathsf{P}_{i'}^1 \in \mathsf{Happy}$ (The set of $\mathsf{Unhappy}$ is determined according to the broadcast in the third step of Com 3, and define a party set Happy, for each P_i^1, if $\mathsf{P}_i^1 \notin \mathsf{Unhappy}$, add P_i^1 to Happy set).

1) Reconstruct $\mathsf{F}^{\mathsf{D}}(X, i, 0, W)$ and $\mathsf{F}^{i'}(i', i, 0, W)$.
2) If for any $\mathsf{P}_i^1 \in \mathsf{Unhappy}$, $\mathsf{Card}\left(\left\{i' \mid \mathsf{P}_{i'}^1 \in \mathsf{Happy} \wedge \mathsf{F}^{\mathsf{D}}(i', i, 0, W) = \mathsf{F}^{i'}(i', i, 0, W)\right\}\right) < 2t + 1$, disqualify the dealer.

Com 4: For each $k \in [n]$, P_k^4 proceeds as follows:

1) For each $\mathsf{P}_i^1 \in \mathsf{Unhappy}$, for each $i' \in [n]$
   - Evaluate the reconstruct value $\mathsf{F}^{\mathsf{D}}(X, i, 0, W)$ at $W = k$ to get $\mathsf{F}^{\mathsf{D}}(X, i, 0, k)$.
   - Set $\mathsf{F}(X, i, 0, k) = \mathsf{F}^{\mathsf{D}}(X, i, 0, k)$.
2) For each $\mathsf{P}_i^1 \in \mathsf{Happy}$, for each $i' \in [n]$,
   - Set $\mathsf{F}(X, i, 0, k) = \mathsf{F}^i(X, i, 0, k)$ received from P_i^1.
3) Invoke Berlekamp-Welch algorithm on $\{\mathsf{F}(0, i, 0, k)\}_{i \in [n]}$ to get $\mathsf{F}(0, 0, 0, k)$. Set $\mathsf{F}(0, 0, 0, k)$ as the share of m.

Fig. 3: FDVSS-1 Protocol

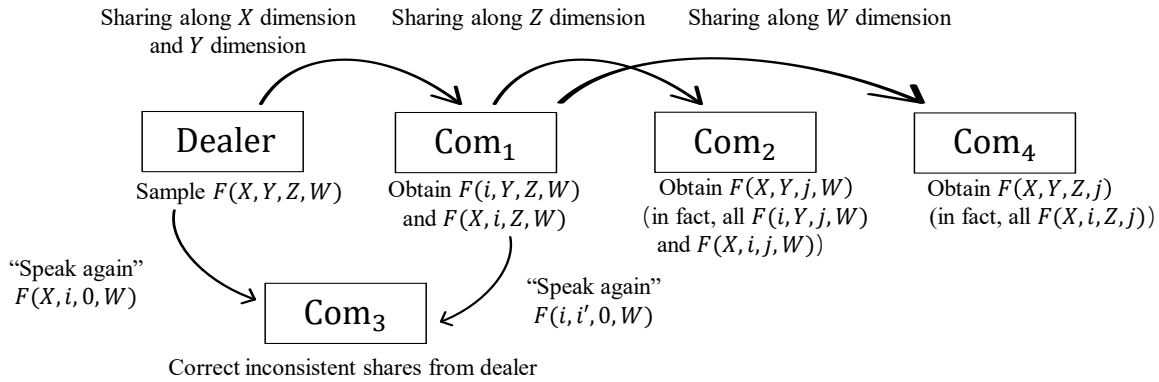

Fig. 4: Communication flow and dimensional usage in the FDVSS-1 protocol

replacing all multi-committee forwarding steps (where $\lambda > 1$) with two new primitives: Future Messaging (Π_{FM}) and Future Commitment (Π_{FC}). These primitives restrict all communication exclusively to the immediate successor committee. We first introduce these two primitives and then present FDVSS-2.

1) Future Messaging: The Future Messaging primitive allows a party in Com_i to deliver a message m to a designated receiver in a distant future committee Com_j, where $j > i$. We realize this long-distance channel by recursively composing single-hop communications ($\lambda = 1$).

For the base case (distance $d = 1$), the message is sent directly via the secure point-to-point channel. For longer distances ($d > 1$), the protocol employs a recursive "share-and-relay" strategy. The sender in Com_i first computes the shares of m using Shamir secret sharing, and then delivers these shares to an intermediate committee, specifically $\mathsf{Com}_{i+\lfloor d/2 \rfloor}$, where the delivery of each share is also a recursive call to Π_{FM}. The parties in this intermediate committee, upon receiving their respective shares, recursively invoke Π_{FM} to forward their shares to the final designated receiver in Com_j. The receiver can then reconstruct m from the relayed shares. This primitive is formally described in Algorithm 1.

Algorithm 1 Future Messaging Π_{FM}.

Input: $\mathsf{S} \in \mathsf{Com}_0$ inputs m.
Output: $\mathsf{R} \in \mathsf{Com}_d$ receives m.

```
S proceeds as follows:
if d = 1 then
    send m to R
else
    sample {[[m]]_1, ..., [[m]]_n} ← Share(m)
    for each j ∈ [n] do
        invoke Π_FM to send [[m]]_j to P^j_⌊d/2⌋
    end for
end if
Each P^j_⌊d/2⌋ ∈ Com_⌊d/2⌋ proceeds as follows:
receive [[m]]_j
invoke Π_FM to send [[m]]_j to R
R proceeds as follows:
receive [[m]]_1, ..., [[m]]_n
m ← Reconst([[m]]_1, ..., [[m]]_n)
```

Lemma 1 (Correctness and Privacy of Future Messaging). *Let* $(\mathsf{Share}, \mathsf{Reconst})$ *be* (t, n) *robust Shamir secret sharing scheme. In protocol* Π_{FM}*, if the sender* $\mathsf{S} \in \mathsf{Com}_0$ *is honest, the receiver* $\mathsf{R} \in \mathsf{Com}_d$ *will correctly receive the message* m *and no information about* m *is revealed to the adversary during the transmission process.*

Proof. Correctness: Since the sender is honest, it always sends the correct message or shares. When $d = 1$, the message is delivered directly to the receiver, thus R obtains correct m. When $d \geq 2$, S splits m into n shares $[\![\mathsf{m}]\!]_1, \ldots, [\![\mathsf{m}]\!]_n$ and sends them to the parties in $\mathsf{Com}_{\lfloor d/2 \rfloor}$. At least $2t+1$ honest parties in $\mathsf{Com}_{\lfloor d/2 \rfloor}$ forward the correct shares to R. Due to the robustness of the secret sharing scheme, even if up to t dishonest parties send wrong shares, R can still reconstruct the correct m.

Privacy: In $\mathsf{Com}_{\lfloor d/2 \rfloor}$, the adversary corrupting t parties receives t shares of m. For each honest $\mathsf{P}^i_{\lfloor d/2 \rfloor}$, t dishonest parties in $\mathsf{Com}_{\lfloor \lfloor d/2 \rfloor \rfloor}$ get t shares of $[\![\mathsf{m}]\!]_i$. Since at least $t+1$ shares are required to reconstruct the secret in the Shamir secret sharing scheme, the adversary can not get $[\![\mathsf{m}]\!]_i$ of honest $\mathsf{P}^i_{\lfloor d/2 \rfloor}$. Therefore, no information about m is revealed to the adversary during the entire transmission process. □

*Complexity Analysis.*The protocol achieves a recursion depth of $\lceil \log d \rceil$ for delivering message m from Com_0 to Com_d, as the distance is halved at each step. Under the "share-and-relay" strategy, each recursive step introduces a branching factor of n, where a single message is partitioned into n shares that trigger independent recursive calls. Consequently, the total communication complexity, in terms of point-to-point transmissions, scales as $O(n^{\lceil \log d \rceil})$.

2) Future Commitment: The Future Commitment primitive is designed to enable a delayed and consistent broadcast, which is often required for follow-up actions such as complaint resolution in a VSS protocol. It is a direct extension of the Π_{FM} protocol.

The key difference lies in the final hop from Com_{d-1} to Com_d.In Future Commitment, parties in Com_{d-1} broadcast their shares, rather than sending them to a specific receiver in Com_d through point-to-point channels. This broadcast mecha-

nism enables all parties in Com_d to execute the reconstruction, thereby guaranteeing consistent reconstruction, meaning that all honest parties in Com_d will reconstruct the identical message.

This primitive is formally described in Algorithm 2.

Algorithm 2 Future Commitment Π_{FC}.

Input: $\mathsf{S} \in \mathsf{Com}_0$ inputs m.
Output: Com_d receives m.

```
S proceeds as follows:
if d = 1 then
    broadcast m
else
    invoke Π_FM to deliver shares of m towards Com_d
end if
For Com_1, ..., Com_{d-2}, proceed as follows:
execute the steps in Π_FM
Com_{d-1} proceeds as follows:
broadcast all data received from Π_FM
Com_d proceeds as follows:
reconstruct m
```

Lemma 2 (Correctness and Privacy of Future Commitment). *In protocol Π_{FC}, if the sender $\mathsf{S} \in \mathsf{Com}_0$ is honest, the message m can only be reconstructed by the receiving committee Com_d and remains hidden from all other committees until then.*

Proof. Correctness: Since the sender is honest, it broadcasts the correct message m when $d = 1$, inputs m to Π_{FM} when $d > 1$. The transmission from Com_0 to Com_{d-1} follows Π_{FM} and thus inherits its correctness property from Lemma 1. Consequently, parties in Com_d can successfully reconstruct m using the shares broadcast by Com_{d-1}.

Privacy: From Lemma 1, even if there are t dishonest parties in each committee from Com_1 to Com_{d-1}, no information about m is revealed during the transmission process. □

Complexity Analysis. The Π_{FC} protocol is essentially identical to Π_{FM}, differing only in the final hop, where it substitutes Π_{FM}'s point-to-point sends with a broadcast. Thus, the point-to-point communication complexity of Π_{FC} is $O(n^{\lceil \log d \rceil})$ as in Π_{FM}. Since the number of data shares transmitted in this final hop is the same in both protocols, the broadcast complexity of Π_{FC} is also $O(n^{\lceil \log d \rceil})$.

3) Construction of FDVSS-2: We substitute the steps that require multi-committee forwarding in fig. 3 as follows:

- In the third step of Com_1, the direct transmission of each $\mathsf{F}(X, i, 0, k)$ destined for P_k^4 in Com_4 is replaced by an invocation of Π_{FM}.
- In the second steps of both the Dealer and Com_1, the procedure of computing and transmitting shares to Com_3 is replaced by a direct invocation of Π_{FC} with Com_4 designated as the receiving committee. And parties in Com_3 only need to broadcast the shares of $\mathsf{F}^{\mathsf{D}}(X, i, 0, W)$ and $\mathsf{F}^{i'}(i', i, 0, W)$ for those $\mathsf{P}_i^1 \in \mathsf{Unhappy}$ and $\mathsf{P}_{i'}^1 \notin \mathsf{Unhappy}$.

4) Security Analysis for FDVSS-2:

Theorem 3. *The FDVSS-2 protocol is a perfectly secure VSS protocol in the fully dynamic model with $\lambda = 1$, satisfying correctness, privacy, and commitment against a Byzantine adversary corrupting up to $t < n/3$ parties in each committee.*

The proof follows from the combined properties of Π_{FM}, Π_{FC}, and the FDVSS-1 protocol. We omit the details here for brevity.

5) Complexity Analysis for FDVSS-2: The dominant communication overhead arises from two components. First, the Dealer utilizes Π_{FC} to transmit n instances of $\mathsf{F}(X, i, 0, W)$ to Com_3, incurring an $O(n^5)$ communication cost. Second, Com_1 sends n^2 instances of $\mathsf{F}(i, i', 0, W)$ to Com_3 via Π_{FC}, and additionally transmits n^2 instances of $\mathsf{F}(X, i, 0, k)$ to Com_4 via Π_{FM}. These steps also result in an $O(n^5)$ overhead. Therefore, the total communication complexity is $O(n^5)$.

IV. Fully Dynamic-Committee Proactive Secret Sharing

We present the Fully Dynamic-Committee Proactive Secret Sharing protocol, which is based on FDVSS-1 and relies on two sub-protocols: Randomness Generation and Redistribution. We begin by introducing these sub-protocols.

A. Randomness Generation

In the Randomness Generation protocol, each party in Com_0 selects an individual random value and share it with Com_4 via FDVSS. As a result, each participant in Com_4 obtains a set of random shares. Critically, the parties in Com_4 use a public Vandermonde matrix to linearly combine the shares of n random values from Com_0, thereby generating $n-t$ globally shared random values.

This protocol is formally described in Algorithm 3.

Algorithm 3 Randomness Generation Π_{RG}.

Public parameters: A Vandermonde matrix $M = Van^{n\times(n-t)}(\beta_1, \cdots, \beta_n)$.
Input: Each $\mathsf{P}_i^0 \in \mathsf{Com}_0$ inputs a random number m_i.
Output: Each $\mathsf{P}_j^4 \in \mathsf{Com}_4$ receives the share of $n-t$ random numbers $r_1, \cdots, r_{n-t}$.

```
Each P_i^0 ∈ Com_0 proceeds as follows:
invoke Π_FDVSS-1 to share m_i with Com_4.
Each P_j^4 ∈ Com_4 proceeds as follows:
receive m_{1,j}, ..., m_{n,j} from Π_FDVSS-1.
compute [r_{1,j}, ..., r_{n-t,j}]^T = M^T [m_{1,j}, ..., m_{n,j}]^T
```

Lemma 3. *The protocol Π_{RG} satisfies the following properties:*

- *Correctness: For $k \in [1, n-t]$, each party in Com_4 holds a valid share of the global secret r_k.*
- *Randomness: The $n-t$ global secrets $r_1, \ldots, r_{n-t}$ defined by the protocol are uniformly random and mutually independent over the underlying field $\mathbb{F}$.*
- *Privacy: The adversary corrupting up to $t < n/3$ parties in each committee learns no information about the $n-t$ global secrets $r_1, \ldots, r_{n-t}$.*

Proof. Correctness: After the execution of the FDVSS, for all $i \in [n]$, each party P_j^4 in Com_4 obtains a valid share of m_i, namely $m_{i,j}$. For each k, linearly combining these shares as $\sum_{i=1}^{n}(\beta_i)^{k-1}m_{i,j}$ yields the share of $r_k = \sum_{i=1}^{n}(\beta_i)^{k-1}m_i$ due to the additive homomorphism property of Shamir secret sharing scheme.

Randomness: Define $\mathbf{m} = [m_1, \ldots, m_n]^T$, $\mathbf{r} = [r_1, \ldots, r_{n-t}]$. In Com_0, let H be the set of $n-t$ honest parties and C be the set of t dishonest parties in Com_0. We have $\mathbf{r} = M_H^T\mathbf{m}_H + M_C^T\mathbf{m}_C$. The matrix M_H^T is a $(n-t)\times(n-t)$ Vandermonde matrix formed by the columns β_i corresponding to H. Since M_H^T is invertible and $\mathbf{m}_H$ is uniformly random, while $\mathbf{m}_C$ is fixed by the adversary, the calculation of $\mathbf{r}$ constitutes a bijective affine transformation of $\mathbf{m}_H$. Therefore, $\mathbf{r}$ is uniformly random in $\mathbb{F}^{n-t}$ and its components are mutually independent.

Privacy: Due to the privacy of FDVSS-1, the adversary corrupting up to t parties in each committee learns nothing about m_i from the honest party P_i^0. For each global secret $r_k = \sum_{i=1}^{n}(\beta_i)^{k-1}m_i$, the adversary holds only at most t shares of it, and thus gains no information about r_k. □

Complexity Analysis. All parties in Com_0 invoke FDVSS-1 in parallel, resulting in a total communication complexity of $O(n^5)$.

B. Redistribution

The idea of the Redistribution protocol is to blind existing shares by adding a specially constructed zero-constant polynomial. In simple terms, the original secret s is shared by a degree-t polynomial $f(x)$ with $f(0) = s$. The protocol introduces a degree-t blinding polynomial $f'(x)$ where $f'(0) = 0$. By setting the new sharing polynomial $g(x) = s + f'(x)$, we ensure the secret s remains the constant term of the new distribution.

The key lies in the distributed construction of $f'(x)$. Consider that each P_i in Com_0 possesses the shares of t random numbers $r_1, \ldots, r_t$. P_i locally constructs a portion of the blinding polynomial,$[f'(x)]_i = \sum_{k=1}^{t} r_{k,i}x^k$, and then combines it with the original share s_i to form a new local polynomial $[g(x)]_i = s_i + [f'(x)]_i$.

After that, P_i evaluates its local polynomial $[g(x)]_i$ at the indices corresponding to the parties in Com_1. Each P_j in Com_1 receives a set of values $\{[g(j)]_1, \ldots, [g(j)]_n\}$ from all parties in Com_0. This enables P_j to obtain its new redistributed share of m, namely $g(j)$.

This protocol is formally described in Algorithm 4.

Lemma 4 (Correctness and Privacy of Redistribution). *In protocol Π_{RD}, if the parties in Com_k hold valid shares of the secret s and t random numbers, then the parties in Com_{k+1} obtain another set of shares of s, and the adversary corrupting t parties in each committee learns no information about s.*

Proof. Correctness: The correctness follows immediately from the protocol construction.

Privacy: For the degree-t polynomial $g(x) = s + \sum_{k=1}^{t} r_k x^k$, the dishonest parties in Com_k hold t shares of each coefficient, so the adversary cannot reconstruct $g(x)$ or s. Similarly, the dishonest parties in Com_{k+1} hold only t points on $g(x)$, which is insufficient for reconstruction. Moreover, since the shares of m in Com_k and Com_{k+1} are generated from two different degree-t polynomials, the adversary learns no information about m from these $2t$ shares. □

Algorithm 4 Redistribution Π_{RD}.

Input: Each $\mathsf{P}_i^k \in \mathsf{Com}_k$ inputs its share of s and t random numbers, s_i and $r_{1,i}, \ldots, r_{t,i}$.
Output: Each $\mathsf{P}_j^{k+1} \in \mathsf{Com}_{k+1}$ receives the new share of m.

Each $\mathsf{P}_i^k \in \mathsf{Com}_k$ **proceeds as follows:**
for $j \in [n]$ **do**
  compute $m_{j,i} = s_i + \sum_{l=1}^{t} r_{l,i} j^l$
  send $m_{j,i}$ to P_j^{k+1}
end for
Each $\mathsf{P}_j^{k+1} \in \mathsf{Com}_{k+1}$ **proceeds as follows:**
receive $m_{j,1}, \ldots, m_{j,n}$ from Com_k.
reconstruct $m_j = \mathsf{Shamir.Reconst}(m_{j,1}, \ldots, m_{j,n})$.
set the new share of s as $s_j = m_j$.

C. FDPSS

We combine the preceding primitives to construct the final Fully Dynamic-Committee Proactive Secret Sharing protocol. This protocol operates in a pipelined manner and consists of two phases:

- Sharing Phase (Com_0 to Com_4):
  1. The Dealer shares the secret s to Com_4 via $\Pi_{\mathsf{FDVSS-1}}$. Simultaneously, Com_0 invokes Π_{RG}.
  2. Com_1 to Com_4 also invoke Π_{RG} to provide future randomness.
  3. Com_4 receives the initial shares of s and redistributes them to Com_5 via Π_{RD} (The randomness for it is provided by Com_0).
- Redistribution Phase(Com_5 onwards):
  1. $\mathsf{Com}_i (i \geq 5)$ receives the share of s that was redistributed by the preceding committee Com_{i-1}.
  2. Com_i then redistributes the shares of s to the next committee Com_{i+1} via Π_{RD} and simultaneously invokes Π_{RG}.

Critically, the efficiency of FDPSS derives from the fully dynamic nature of our VSS. This specific property enables a pipelined architecture, which amortizes the DPSS hand-off overhead, effectively reducing the communication to a single round.

1) Security Analysis:

Theorem 4. *The FDPSS protocol satisfies the following two properties:*

- *Integrity: If the dealer is honest and the adversary corrupts at most $t < n/3$ parties in each committee, the shares of honest parties in $\mathsf{Com}_i (i \geq 4)$ can be correctly computed and the secret s can be reconstructed from these shares.*
- *Secrecy: If the adversary corrupts at most t parties in each committee, it learns no information about the secret s.*

According to the FDVSS-1 protocol and the Redistribution protocol, both properties hold trivially. We omit the proof.

2) Complexity Analysis: The highest inter-committee communication complexity stems from the Π_{RG} primitive, which is $O(n^5)$.

V. Evaluation

In this section, we evaluate the performance of our proposed FDVSS-1 and FDPSS protocols. Our experiments focus on quantifying the computational efficiency and communication overhead across various committee sizes.

A. Experimental Setup

We implemented FDVSS-1 and FDPSS in Go (version 1.17.6). Leveraging the property that perfectly secure schemes rely solely on guessing probability, we instantiated our protocols using the native uint64 type ($p \approx 2^{64}$). This choice ensures practical security (2^{-64}) while enabling highly efficient CPU-native arithmetic, avoiding the overhead of large fields (e.g., 256-bit) required by computational baselines. Additionally, we applied standard memory and vectorization optimizations to minimize the overhead of high-dimensional polynomial operations.

All experiments were conducted on a consumer-grade laptop equipped with an Intel Core i5-1135G7 processor (@2.40GHz) and 64GB of RAM, running Ubuntu 20.04 LTS. The results report the computation time and message length of the first party in each committee, with the average values taken from 10 independent executions. In our experiments, all parties are assumed to be honest; therefore, no accusations or dispute resolution mechanisms were triggered.

B. Performance of FDVSS-1

Table III presents the comprehensive performance of our protocol across different committee sizes.

1) Small-Scale Efficiency ($n \leq 32$): In this range, FDVSS-1 is efficiently executable. At $n = 32$, the protocol completes the entire computation spanning four committees in 26 ms. This efficiency is attributed to our design's exclusive reliance on lightweight field arithmetic, which avoids the computational overhead of complex cryptographic primitives. Regarding communication, despite the $O(n^4)$ complexity, the traffic volume remains low at approximately 1.06 MB for dealer when $n = 32$, remaining well within the capacity of modern network environments.

2) Scalability and Feasibility ($n \geq 64$): As n scales, the impact of the polynomial computational complexity in FDVSS-1 becomes the dominant factor. For instance, at $n = 256$, the total computation time increases to approximately 580 seconds. In terms of communication, the total traffic per dealer approaches 4 GB at $n = 256$. Distributing this volume over a typical 100 Mbps WAN link would incur a transmission delay of roughly 300 seconds, which is comparable in magnitude to the computational overhead. While substantial, this total latency remains within the realm of feasibility for high-value distributed applications where perfect security is prioritized over real-time performance.

C. Performance of FDPSS

For the hand-off phase of the FDPSS protocol, we evaluate the same performance metrics as for FDVSS, and the results are provided in Table IV.

The experimental results mirror the trends observed in FDVSS. For committee sizes $n \leq 32$, the protocol demonstrates efficient performance. However, at $n = 128$, the total traffic per node reaches approximately 1 GB. This substantial overhead is attributed to the pipelined architecture of FDPSS: during a single hand-off phase, each node is required to simultaneously fulfill the role of a Dealer to initiate an FDVSS instance, while also serving as a participant across different committees in other protocols.

Similarly, while FDPSS incurs higher overhead in large-scale clusters, this trade-off is justified by its architectural robustness. By eliminating both the quiescent window and the reliance on trusted setups, the protocol is uniquely suited for permissionless systems where security and decentralization are prioritized over raw scalability.

VI. Conclusion

In this paper, we identified and eliminated the "quiescent window" limitation in synchronous DPSS schemes. We proposed the Fully Dynamic-Committee model and instantiated it with a suite of Fully Dynamic Verifiable Secret Sharing protocols. Our main contribution, FDVSS-1, achieves perfect security with near-optimal round complexity (4 rounds) and $O(n^4)$ communication. Furthermore, we presented FDPSS, which enables share redistribution in a single round, thereby removing the requirement for a static committee during hand-offs. Our protocols distinguish themselves by achieving perfect security and optimal resilience ($t < n/3$) without relying on any trusted setup or computational assumptions. While our current FDPSS construction incurs an $O(n^5)$ communication overhead to ensure these strong security properties, it provides a feasibility baseline for fully dynamic secret sharing. Future work includes exploring techniques to further reduce the communication complexity of fully dynamic schemes and extending these paradigms to asynchronous network settings.

Acknowledgments

This work is supported by the Key R&D Program of Shandong Province, China(2024ZLGX05), the National Natural Science Foundation of China (72334003).

	Dealer		Com1		Com2		Com3	Com4
n	Time	Size	Time	Size	Time	Size	Time	Time
8	0.28 ms	6.7 KB	0.08 ms	5.08 KB	0.02 ms	0.13 KB	0.04 ms	0.06 ms
16	2.1 ms	92.6 KB	0.16 ms	28.9 KB	0.08 ms	0.37 KB	0.10 ms	0.38 ms
32	23.4 ms	1.06 MB	1.25 ms	155 KB	0.21 ms	1.18 KB	0.22 ms	1.78 ms
64	550 ms	16.6 MB	9.55 ms	0.98 MB	1.22 ms	4.3 KB	1.9 ms	9.39 ms
128	12.6 s	246 MB	119 ms	6.64 MB	9.76 ms	16.6 KB	9.02 ms	43.8 ms
256	334 s	3.95 GB	1.91 s	108 MB	77.4 ms	66.4 KB	72 ms	174 ms

TABLE III: Performance of FDVSS-1 with $t = \lfloor \frac{n-1}{3} \rfloor$. Due to the all-honest setting, parties of Com_3 are not required to broadcast any shares; therefore, its communication overhead is not recorded. Similarly, since Com_4 serves as the shareholder committee, its communication overhead is also excluded.

n	Time	Size
8	5.57 ms	47.8 KB
16	62 ms	538 KB
32	0.41 s	5.74 MB
64	6.54 s	76.4 MB
128	37.9 s	1.02 GB

TABLE IV: Performance of the hand-off phase of FDPSS protocol with $t = \lfloor \frac{n-1}{3} \rfloor$.

References

[1] A. Shamir, "How to share a secret," *Communications of the ACM*, vol. 22, no. 11, pp. 612–613, 1979.

[2] A. Herzberg, S. Jarecki, H. Krawczyk, and M. Yung, "Proactive secret sharing or: How to cope with perpetual leakage," in *annual international cryptology conference*. Springer, 1995, pp. 339–352.

[3] D. A. Schultz, B. Liskov, and M. Liskov, "Mobile proactive secret sharing," in *Proceedings of the twenty-seventh ACM symposium on Principles of distributed computing*, 2008, pp. 458–458.

[4] S. K. D. Maram, F. Zhang, L. Wang, A. Low, Y. Zhang, A. Juels, and D. Song, "Churp: dynamic-committee proactive secret sharing," in *Proceedings of the 2019 ACM SIGSAC Conference on Computer and Communications Security*, 2019, pp. 2369–2386.

[5] T. M. Wong, C. Wang, and J. M. Wing, "Verifiable secret redistribution for archive systems," in *First International IEEE Security in Storage Workshop, 2002. Proceedings.* IEEE, 2002, pp. 94–105.

[6] V. Goyal, A. Kothapalli, E. Masserova, B. Parno, and Y. Song, "Storing and retrieving secrets on a blockchain," in *IACR International Conference on Public-Key Cryptography*. Springer, 2022, pp. 252–282.

[7] J. Baron, K. E. Defrawy, J. Lampkins, and R. Ostrovsky, "Communication-optimal proactive secret sharing for dynamic groups," in *International conference on applied cryptography and network security*. Springer, 2015, pp. 23–41.

[8] Y. Desmedt and S. Jajodia, "Redistributing secret shares to new access structures and its applications," Technical Report ISSE TR-97-01, George Mason University, Tech. Rep., 1997.

[9] C. Gentry, S. Halevi, H. Krawczyk, B. Magri, J. B. Nielsen, T. Rabin, and S. Yakoubov, "Yoso: You only speak once: Secure mpc with stateless ephemeral roles," in *Annual International Cryptology Conference*. Springer, 2021, pp. 64–93.

[10] A. R. Choudhuri, A. Goel, M. Green, A. Jain, and G. Kaptchuk, "Fluid mpc: secure multiparty computation with dynamic participants," in *Annual International Cryptology Conference*. Springer, 2021, pp. 94–123.

[11] D. Dolev, C. Dwork, O. Waarts, and M. Yung, "Perfectly secure message transmission," *J. ACM*, vol. 40, no. 1, p. 17–47, Jan. 1993. [Online]. Available: https://doi.org/10.1145/138027.138036

[12] M. Ben-Or, S. Goldwasser, and A. Wigderson, "Completeness theorems for non-cryptographic fault-tolerant distributed computation (extended abstract)," in *20th ACM STOC*. ACM Press, May 1988, pp. 1–10.

[13] R. Gennaro, Y. Ishai, E. Kushilevitz, and T. Rabin, "The round complexity of verifiable secret sharing and secure multicast," in *Proceedings of the thirty-third annual ACM symposium on Theory of computing*, 2001, pp. 580–589.

[14] J. Katz, C.-Y. Koo, and R. Kumaresan, "Improving the round complexity of vss in point-to-point networks," in *International Colloquium on Automata, Languages, and Programming*. Springer, 2008, pp. 499–510.

[15] B. Applebaum, E. Kachlon, and A. Patra, "The round complexity of perfect mpc with active security and optimal resiliency," in *2020 IEEE 61st Annual Symposium on Foundations of Computer Science (FOCS)*. IEEE, 2020, pp. 1277–1284.

[16] A. Bienstock, D. Escudero, and A. Polychroniadou, "On linear communication complexity for (maximally) fluid mpc," in *Annual International Cryptology Conference*. Springer, 2023, pp. 263–294.

[17] ——, "Perfectly secure fluid mpc with abort and linear communication complexity," *IACR Communications in Cryptology*, vol. 1, no. 4, 2025.

[18] L. Braun, I. Damgård, and C. Orlandi, "Secure multiparty computation from threshold encryption based on class groups," in *CRYPTO 2023, Part I*, ser. LNCS, H. Handschuh and A. Lysyanskaya, Eds., vol. 14081. Springer, Cham, Aug. 2023, pp. 613–645.

[19] S. Kolby, D. Ravi, and S. Yakoubov, "Constant-round yoso mpc without setup," *IACR Communications in Cryptology*, vol. 1, no. 3, 2024.

[20] B. David, A. Mondal, and R. Satish, "Rumors mpc: God for dynamic committees low communication via constant-round chat," *Cryptology ePrint Archive*, 2025.

[21] Y. Zeng, K. Yang, D. Feng, and M. Zhang, "Ion: Concretely efficient submaximal-fluid mpc with linear communication," *Cryptology ePrint Archive*, 2025.

[22] W. Yu, M. Xu, B. Wu, S. Duan, and X. Cheng, "Ad-mpc: Fully asynchronous dynamic mpc with guaranteed output delivery," *Cryptology ePrint Archive*, 2024.

[23] J. B. Nielsen, J. Ribeiro, and M. Obremski, "Public randomness extraction with ephemeral roles and worst-case corruptions," in *Annual International Cryptology Conference*. Springer, 2022, pp. 127–147.

[24] C.-D. Liu-Zhang, E. Masserova, J. Ribeiro, P. Soni, and S. A. Thyagarajan, "Improved yoso randomness generation with worst-case corruptions," in *International Conference on Financial Cryptography and Data Security*. Springer, 2024, pp. 73–89.

[25] B. David, G. Deligios, A. Goel, Y. Ishai, A. Konring, E. Kushilevitz, C.-D. Liu-Zhang, and V. Narayanan, "Perfect mpc over layered graphs," in *Annual International Cryptology Conference*. Springer, 2023, pp. 360–392.

[26] G. Deligios, A. Konring, C. Liu-Zhang, and V. Narayanan, "Statistical layered MPC," in *TCC (4)*, ser. Lecture Notes in Computer Science, vol. 15367. Springer, 2024, pp. 362–394.

[27] M. Ito, A. Saito, and T. Nishizeki, "Secret sharing scheme realizing general access structure," *Electronics and Communications in Japan (Part III: Fundamental Electronic Science)*, vol. 72, no. 9, pp. 56–64, 1989.

[28] A. Shamir, "How to share a secret," *Communications of the Association for Computing Machinery*, vol. 22, no. 11, pp. 612–613, Nov. 1979.

[29] E. R. Berlekamp and M. E. Welzl, "Error correction of algebraic block codes." *US Patent*, no. 4,633,470, 1986.

[30] G. Asharov and Y. Lindell, "A full proof of the BGW protocol for perfectly secure multiparty computation," *Journal of Cryptology*, vol. 30, no. 1, pp. 58–151, 2017. [Online]. Available: https://doi.org/10.1007/s00145-015-9214-4

Appendix
Proof of FDVSS-1

We first present the algebraic claims required for the proof. These claims establish the uniqueness of multivariate polynomial interpolation used in our consistency checks.

Claim 1. *Let t be a nonnegative integer, let $\alpha_1, \ldots, \alpha_{t+1}$ be $t+1$ distinct elements in $\mathbb{F}$, and let $f_1(x,y), \ldots, f_{t+1}(x,y)$ be $t+1$ bivariate polynomials of degree t in each variable. Then, there exists a unique trivariate polynomial $S(x,y,z)$ of degree t such that for every $k = 1, \ldots, t+1$ it holds that $S(x,y,\alpha_k) = f_k(x,y)$.*

Proof. This is an extension of Claim 5.2 in [30]. We prove the existence by construction using Lagrange interpolation. For $k \in \{1, \ldots, t+1\}$, define the k-th Lagrange basis polynomial for the set $\{\alpha_1, \ldots, \alpha_{t+1}\}$:

$$L_k(z) = \prod_{j=1, j\neq k}^{t+1} \frac{z - \alpha_j}{\alpha_k - \alpha_j}$$

There is a polynomial S satisfies the required degree constraint.

$$S(x,y,z) = \sum_{k=1}^{t+1} f_k(x,y) \cdot L_k(z)$$

The interpolation condition $S(x,y,\alpha_k) = f_k(x,y)$ holds because $L_j(\alpha_k) = \delta_{jk}$ (the Kronecker delta).

To prove uniqueness, assume $S'(x,y,z)$ is another polynomial satisfying the conditions. Let $D(x,y,z) = S(x,y,z) - S'(x,y,z)$. However, for all $t+1$ distinct values α_k, $D(x,y,\alpha_k) = S(x,y,\alpha_k) - S'(x,y,\alpha_k) = f_k(x,y) - f_k(x,y) = 0$. A polynomial in z of degree at most t with $t+1$ distinct roots must be the zero polynomial. Thus, $D(x,y,z) \equiv 0$, which implies $S = S'$. □

Claim 2. *Let t be a nonnegative integer, let $K \subseteq [n]$ be a set of indices such that $|K| \geq t+1$, and let $\{\alpha_k\}_{k\in K}$ be distinct nonzero elements in $\mathbb{F}$. Let $\{f_k(x,z)\}_{k\in K}$ and $\{g_k(y,z)\}_{k\in K}$ be sets of degree-t bivariate polynomials. If for every $i,j \in K$, it holds that $f_i(\alpha_j, z) = g_j(\alpha_i, z)$, then there exists a unique trivariate polynomial $S(x,y,z)$ of degree t, such that $f_k(x,z) = S(x,\alpha_k,z)$ and $g_k(y,z) = S(\alpha_k,y,z)$ for every $k \in K$.*

Proof. This is an extension of Claim 5.3 in [30]. Let $L \subseteq K$ be any subset with $|L| = t+1$. By Claim 1, there exists a unique polynomial $S(x,y,z)$ with individual degrees at most t such that $S(x,\alpha_l,z) = f_l(x,z)$ for all $l \in L$. We now prove $g_k(y,z) = S(\alpha_k,y,z)$ holds for all $k \in K$. For any $k \in K, l \in L$, we have $g_k(\alpha_l,z) = f_l(\alpha_k,z)$ and $f_l(\alpha_k,z) = S(\alpha_k,\alpha_l,z)$. Since both $g_k(y,z)$ and $S(\alpha_k,y,z)$ are degree-t in y, the equality $g_k(y,z) \equiv S(\alpha_k,y,z)$ holds for all $k \in K$. It remains to prove that $f_k(x,z) = S(x,\alpha_k,z)$ holds for all $k \in K$. For all $k \in L$, this is true by the construction of S. For all $k \in K \setminus L$, the polynomials $f_k(x,z)$ and $S(x,\alpha_k,z)$ are equal at $t+1$ points $\{\alpha_j\}_{j\in L}$ since $f_k(\alpha_j,z) = g_j(\alpha_k,z)$ and $S(\alpha_j,\alpha_k,z) = g_j(\alpha_k,z)$. Thus, $f_k(x,z) \equiv S(x,\alpha_k,z)$ for all $k \in K$. The uniqueness of S follows from its unique construction via L. □

The following two lemmas ensure that if the dealer is honest, then honest parties in Com_1 will be happy, while any dishonest party distributing inconsistent shares for comparison will be detected and marked as $\mathsf{Unhappy}$. The correctness of FDVSS-1 is based on these lemmas.

Lemma 5. *If the dealer D is honest, all honest parties in Com_1 will not be marked as $\mathsf{Unhappy}$.*

Proof. If D is honest, the shares it sends to Com_1 are all correct. Let H_1 be the index set of the honest parties in Com_1. All honest parties in Com_1 will execute the protocol correctly. Therefore, they are guaranteed to form a clique of size $2t+1$ in the consistency graph G_j broadcast by every honest party P_j^2 in Com_2. Even if dishonest parties in Com_2 broadcast wrong consistency graphs, they cannot exclude H_1 from the set I. So, all honest parties in Com_1 will not be $\mathsf{Unhappy}$. □

Lemma 6. *If the dealer D is honest, any dishonest P_i^1 that uses an incorrect $\mathsf{F}(X,i,Z,W)$ or $\mathsf{F}(i,Y,Z,W)$ during Step 1) for comparison must be marked as $\mathsf{Unhappy}$.*

Proof. That is, if D is honest, a dishonest P_i^1 will be added to the set I only if it uses the correct $\mathsf{F}(X,i,Z,W)$ and $\mathsf{F}(i,Y,Z,W)$ from D. A party is added to I only if it appears in the clique of at least $2t+1$ parties from Com_2, and let H_2 be the index set of honest parties among them, where $|H_2| \geq t+1$. For each $j \in H_2$, the clique of G_j must contain H_1. P_i^1 must be consistent with all the honest parties in Com_1 across all $j \in H_2$ so that i can be included in I. This means, for each $i' \in H_1$ and $j \in H_2$, the following must hold:

$$\mathsf{F}^i(i',i,j,W) = \mathsf{F}^{i'}(i',i,j,W), \mathsf{F}^i(i,i',j,W) = \mathsf{F}^{i'}(i,i',j,W)$$

We now prove $\mathsf{F}^i(X,i,Z,W)$ is equal to $\mathsf{F}(X,i,Z,W)$ from D. For each $i' \in H_1$, $\mathsf{P}_{i'}^1$ uses the correct polynomial from D, so

$$\mathsf{F}^i(i',i,j,W) = \mathsf{F}(i',i,j,W) \quad \text{for each } i' \in H_1$$

Given that $\mathsf{F}^i(X,i,j,W)$ and $\mathsf{F}(X,i,j,W)$ are polynomials of degree at most t in X, the two polynomials must be identical since they agree on $|H_1|(|H_1| \geq 2t+1)$ points. Therefore, for each $j \in H_2$:

$$\mathsf{F}^i(X,i,j,W) = \mathsf{F}(X,i,j,W)$$

According to Claim 1, there exists a unique polynomial $G(X,i,Z,W)$ such that $G(X,i,j,W) = \mathsf{F}^i(X,i,j,W)$ for all $j \in H_2$. Since $\mathsf{F}^i(X,i,j,W) = \mathsf{F}(X,i,j,W)$ for all $j \in H_2$, and $\mathsf{F}(X,i,Z,W)$ is also a degree-t polynomial in Z, it must be that $G(X,i,Z,W) = \mathsf{F}(X,i,Z,W)$. Therefore, $\mathsf{F}^i(X,i,Z,W) = \mathsf{F}(X,i,Z,W)$. By the same logic, $\mathsf{F}^i(i,Y,Z,W) = \mathsf{F}(i,Y,Z,W)$. This concludes that the dishonest P_i^1 must use polynomials identical to correct polynomials from D; otherwise, it will fail the consistency check, resulting in its exclusion from I and be marked as $\mathsf{Unhappy}$. □

Theorem 2. The FDVSS-1 protocol is a perfectly secure verifiable secret sharing scheme in the fully dynamic model. Specifically, assuming $n \geq 3t+1$, it satisfies the properties

of correctness, commitment, and privacy against a Byzantine adversary that corrupts up to t parties in each committee.

Proof. Correctness: The correctness of the protocol is based on the above lemmas. If D is honest, it sends correct shares to Com_1 and Com_3. For an honest P_i^1, it will share the correct $\mathsf{F}(i', i, 0, W)$ with Com_3 and send the correct $\mathsf{F}(X, i, 0, k)$ P_k^4 in Com_4. Moreover, P_i^1 is guaranteed to be Happy, according to Lemma 5. For a dishonest $\mathsf{P}_{i'}^1$, if $\mathsf{P}_{i'}^1$ uses an incorrect polynomial for comparison, it will be unhappy according to Lemma 6. Due to the robustness of Shamir secret sharing, dishonest parties in Com_3 broadcasting incorrect shares do not affect the reconstruction of these polynomials, thus $\mathsf{F}^{\mathsf{D}}(X, i', 0, W)$ will pass the check. By now, P_k^4 holds the $\mathsf{F}^{\mathsf{D}}(X, i', 0, W)$ for those unhappy $\mathsf{P}_{i'}^1$ and $\mathsf{F}(X, i, 0, k)$ for those honest and happy P_i^1. $\mathsf{F}(0, 0, 0, k)$ can be correctly recovered even if happy but dishonest P_i^1 send incorrect $\mathsf{F}(X, i, 0, j)$. Furthermore, even if a dishonest party in Com_2 broadcasts an incorrect consistency graph, the set I will not be affected. This is because, a party P_j^1 using incorrect polynomials will not be added to I according to Lemma 6, while for a party $\mathsf{P}_{i'}^1$ using the correct polynomials, i' is guaranteed to be in the cliques of at least $2t+1$ honest parties in Com_2 and then i' will be added to I. Therefore, during the reconstruction phase, honest parties in Com_4 broadcast the correct $\mathsf{F}(0, 0, 0, j)$. Even if other dishonest parties broadcast incorrect shares, the secret m will still be correctly reconstructed.

Privacy: The polynomials $\mathsf{F}(i, Y, Z, W)$ and $\mathsf{F}(X, i, Z, W)$ received by P_i^1 in Com_1 can be viewed as two shares of m: $\mathsf{F}(i, 0, 0, 0)$ and $\mathsf{F}(0, i, 0, 0)$. Similarly, parties in Com_2 receive $\mathsf{F}(0, 0, j, 0)$, and parties in Com_4 receive $\mathsf{F}(0, 0, 0, j)$. These values represent shares along different dimensions. For a single dimension (for example, the X-dimension), $t+1$ values of $\mathsf{F}(X, 0, 0, 0)$ are required to reconstruct $\mathsf{F}(0, 0, 0, 0)$, i.e., the secret m. However, the adversary holds at most t values from the dishonest parties. Furthermore, the adversary corrupting t parties in Com_3 receives t shares of $\mathsf{F}(X, i, 0, W)$ (or $\mathsf{F}(i', i, 0, W)$), which is not helpful for extracting information about the secret m.

Commitment: If D is not disqualified, it implies that there are at least $2t+1$ happy parties in Com_1. Let H_1' be the index set of the honest parties among them, where $|H_1'| \geq t+1$. H_1' is contained in the clique of at least $2t+1$ parties in Com_2. Let H_2' be the index set of the honest parties among these parties, where $|H_2'| \geq t+1$. For each $\mathsf{P}_j^2 \in H_2'$, it must hold that $\mathsf{F}^i(i', i, j, W) = \mathsf{F}^{i'}(i', i, j, W)$ and $\mathsf{F}^i(i, i', j, W) = \mathsf{F}^{i'}(i, i', j, W)$ for any pair $i, i' \in H_1'$. According to Claim 2, the polynomials of the parties in H_1' originate from a unique trivariate t-degree polynomial $G(X, Y, j, W)$. Based on an extension of Claim 1, the set of polynomials $\{G(X, Y, j, W)\}_{j \in H_2'}$ originates from a unique quadrivariate t-degree polynomial $G(X, Y, Z, W)$. For each unhappy party P_i^1, it holds that $\mathsf{F}^{\mathsf{D}}(i', i, 0, W) = \mathsf{F}^{i'}(i', i, 0, W)$ for at least $2t+1$ parties $\mathsf{P}_{i'}^1 \in H_1'$. Since $\mathsf{P}_{i'}^1$ is honest, $\mathsf{F}^{i'}(i', i, 0, W) = G(i', i, 0, W)$. Given that $\mathsf{F}^{\mathsf{D}}(X, i, 0, W)$ and $G(X, i, 0, W)$ are both t-degree polynomials in X and they agree on $|H_1'|(|H_1'| \geq t+1)$ points, they must be identical. Regardless of what those happy but dishonest parties in Com_1 send to P_k^4, P_k^4 can still correctly reconstruct $G(0, 0, 0, k)$ using the polynomials from other parties since $|H_1' \cup \mathsf{Unhappy}| \geq 2t+1$. At this point, the reconstruction phase is guaranteed to reconstruct a unique value s, where $s = G(0, 0, 0, 0)$. □

Jiarui Li received his M.S. degree in Cyber Science and Technology from Shandong University, Qingdao, China, in 2024. He is currently pursuing a ph.D degree with the School of Cyber Science and Technology at Shandong University, Qingdao, China. His current research interests include blockchain consensus and secure multi-party computation.

Mengzhen Zou received her bachelor's degree in Information Security from China University of Geosciences, Wuhan, China, in 2024. She is currently pursuing a master's degree with the School of Cyber Science and Technology at Shandong University, Qingdao, China. Her current research interests include secure multi-party computation.

Chen Qian received his ph.D degree from University of Rennes /ENS Lyon, France, in 2019. From 2019 to 2022,he worked as a Post-Doc researcher at Norwegian University of Science and Technology,Norway. He is currently a professor with the School of Cyber Science and Technology at Shandong University, Qingdao, China. His current research interests include cryptography, complexity theory and theoretical computer science.

Guoyan Zhang received her Ph.D. in Information Security from Shandong University, Jinan, Shandong, China, in 2008. She is currently a professor with the School of Cryptologic Science and Engineering at Shandong University, Jinan, China. Her current research interests include privacy computing, AI security.